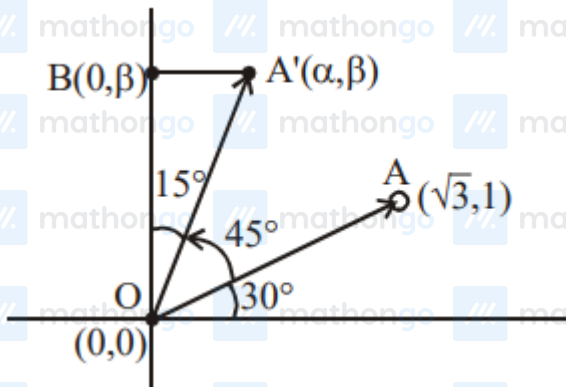


Q1 (1)



$$\begin{aligned}\text{Area of } \Delta (OA'B) &= \frac{1}{2} OA' \cos 15^\circ \times OA' \sin 15^\circ \\ &= \frac{1}{2} (OA')^2 \frac{\sin 30^\circ}{2} \\ &= (3+1) \times \frac{1}{8} = \frac{1}{2}\end{aligned}$$

Q2 (2)

$$\vec{r} \times \vec{a} = \vec{b} \times \vec{r} \Rightarrow \vec{r} \times (\vec{a} + \vec{b}) = 0$$

$$\vec{r} = \lambda(\vec{a} + \vec{b}) \Rightarrow \vec{r} = \lambda(\hat{i} + 2\hat{j} - 3\hat{k} + 2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\vec{r} = \lambda(3\hat{i} - \hat{j} + 2\hat{k}) \dots (1)$$

$$\vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3$$

$$\text{Put } \vec{r} \text{ from (1)} \alpha\lambda = 1 \dots (2)$$

$$\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1$$

$$\text{Put } \vec{r} \text{ from (1)} 2\lambda\alpha - \lambda = 1 \dots (3)$$

Solve (2)&(3)

$$\alpha = 1, \lambda = 1$$

$$\Rightarrow \vec{r} = 3\hat{i} - \hat{j} + 2\hat{k}$$

$$|\vec{r}|^2 = 14 \& \alpha = 1$$

$$\alpha + |\vec{r}|^2 = 15$$

Q3 (28)

Hints and Solutions

MathonGo

$$\vec{c} = \lambda(\vec{a} \times \vec{b})$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix}$$

$$(\vec{a} \times \vec{b}) = 3\hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = \lambda(3\hat{i} - 2\hat{j} + \hat{k}) \cdot (\hat{i} + \hat{j} + 3\hat{k})$$

$$\Rightarrow \lambda(4) = 8 \Rightarrow \lambda = 2$$

$$\vec{c} = 2(\vec{a} \times \vec{b})$$

$$\vec{c} \cdot (\vec{a} \times \vec{b}) = 2|\vec{a} \times \vec{b}|^2 = 28$$

Q4 (1)

$$\vec{r} \times \vec{a} - \vec{r} \times \vec{b} = 0$$

$$\Rightarrow \vec{r} \times (\vec{a} - \vec{b}) = 0$$

$$\Rightarrow \vec{r} = \lambda(\vec{a} - \vec{b})$$

$$\Rightarrow \vec{r} = \lambda(-5\hat{i} - 4\hat{j} + 10\hat{k})$$

$$\text{Also } \vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -3$$

$$\Rightarrow \lambda(-5 - 8 + 10) = -3$$

$$\lambda = 1$$

$$\text{Now } \vec{r} = -5\hat{i} - 4\hat{j} + 10\hat{k}$$

$$= \vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k})$$

$$= -10 + 12 + 10 = 12$$

Q5 (2)

$$\vec{a} \cdot \vec{b} = 1 \Rightarrow -\alpha\beta - \alpha\beta - 3 = 1$$

$$\Rightarrow -2\alpha\beta = 4 \Rightarrow \alpha\beta = -2$$

$$\vec{b} \cdot \vec{c} = -3 \Rightarrow -\beta + 2\alpha + 1 = -3$$

$$\text{Solving (1) \& (2), } (\alpha, \beta) = (-1, 2)$$

Hints and Solutions

MathonGo

$$\frac{1}{3}[\vec{a} \vec{b} \vec{c}] = \frac{1}{3} \begin{vmatrix} \alpha & \beta & 3 \\ -\beta & -\alpha & -1 \\ 1 & -2 & -1 \end{vmatrix}$$

$$= \frac{1}{3} \begin{vmatrix} -1 & 2 & 3 \\ -2 & 1 & -1 \\ 1 & -2 & -1 \end{vmatrix}$$

$$= \frac{1}{3} \begin{vmatrix} 0 & 0 & 2 \\ -2 & 1 & -1 \\ 1 & -2 & -1 \end{vmatrix} = \frac{1}{3}[2(4-1)] = 2$$

Q6 (2)

$$\vec{OP} \perp \vec{OQ}$$

$$\Rightarrow -x + 2y - 3x = 0$$

$$\Rightarrow y = 2x$$

$$|\vec{PQ}|^2 = 20$$

$$\Rightarrow (x+1)^2 + (y-2)^2 + (1+3x)^2 = 20$$

$$\Rightarrow x = 1$$

$\vec{OP}, \vec{OQ}, \vec{OR}$ are coplanar.

$$\Rightarrow \begin{vmatrix} x & y & -1 \\ -1 & 2 & 3x \\ 3 & z & -7 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 2 & -1 \\ -1 & 2 & 3 \\ 3 & z & -7 \end{vmatrix} = 0$$

$$\Rightarrow 1(-14-3z) - 2(7-9) - 1(-z-6) = 0$$

$$\Rightarrow z = -2$$

$$\therefore x^2 + y^2 + z^2 = 1 + 4 + 4 = 9$$

Q7 (486)

$$\text{Let } \vec{x} = \lambda \vec{a} + \mu \vec{b} \quad (\lambda \text{ and } \mu \text{ are scalars})$$

$$\vec{x} = \hat{i}(2\lambda + \mu) + \hat{j}(2\mu - \lambda) + \hat{k}(\lambda - \mu)$$

$$\text{Since } \vec{x} \cdot (3\hat{i} + 2\hat{j} - \hat{k}) = 0$$

$$3\lambda + 8\mu = 0 \quad \dots$$

$$\text{Also Projection of } \vec{x} \text{ on } \vec{a} \text{ is } \frac{17\sqrt{6}}{2}$$

$$\frac{\vec{x} \cdot \vec{a}}{|\vec{a}|} = \frac{17\sqrt{6}}{2}$$

$$6\lambda - \mu = 51$$

$$\text{From (1) and (2) } \lambda = 8, \mu = -3$$

$$\vec{x} = 13\hat{i} - 14\hat{j} + 11\hat{k}$$

$$|\vec{x}|^2 = 486$$

Q8 (4)

$$\vec{a}_{\text{Old}} = 3p\hat{i} + \hat{j}$$

$$\vec{a}_{\text{New}} = (p+1)\hat{i} + \sqrt{10}\hat{j}$$

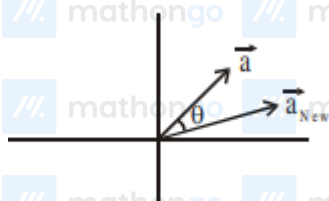
$$\Rightarrow |\vec{a}_{\text{old}}| = |\vec{a}_{\text{New}}|$$

$$\Rightarrow ap^2 + 1 = p^2 + 2p + 1 + 10$$

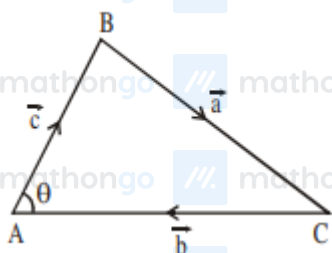
$$8p^2 - 2p - 10 = 0$$

$$4p^2 - p - 5 = 0$$

$$(4p - 5)(p + 1) = 0 \rightarrow p = \frac{5}{4}, -1$$



Q9 (2)



$$|\vec{a}| = 8, |\vec{b}| = 7, |\vec{c}| = 10$$

$$\cos \theta = \frac{|\vec{b}|^2 + |\vec{c}|^2 - |\vec{a}|^2}{2|\vec{b}||\vec{c}|} = \frac{17}{28}$$

Projection of $\vec{c}$ on $\vec{b}$

$$= |\vec{c}| \cos \theta$$

$$= 10 \times \frac{17}{28}$$

$$= \frac{85}{14}$$

Q10 (2)

$$|\vec{a}| = |\vec{b}|, |\vec{a} \times \vec{b}| = |\vec{a}|, \vec{a} \perp \vec{b}$$

$$|\vec{a} \times \vec{b}| = |\vec{a}| \Rightarrow |\vec{a}| |\vec{b}| \sin 90^\circ = |\vec{a}| \Rightarrow |\vec{b}| = 1 = |\vec{a}|$$

$\vec{a}$ and $\vec{b}$ are mutually perpendicular unit

vectors.

$$\text{Let } \vec{a} = \hat{i}, \vec{b} = \hat{j} \Rightarrow \vec{a} \times \vec{b} = \hat{k}$$

$$\cos \theta = \frac{(\hat{i} + \hat{j} + \hat{k}) \cdot \hat{i}}{\sqrt{3} \sqrt{1}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

Let a vector $\alpha\hat{i} + \beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i} + \hat{j}$ by an angle 45° about the origin in counterclockwise direction in the first quadrant. Then the area of triangle having vertices (α, β) , $(0, \beta)$ and $(0, 0)$ is equal to

- (1) $\frac{1}{2}$
- (2) 1
- (3) $\frac{1}{\sqrt{2}}$
- (4) $2\sqrt{2}$

Q2: 16 March (Shift 2) - Single Correct

Let $\vec{a} = \hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = 2\hat{i} - 3\hat{j} + 5\hat{k}$. If $\vec{r} \times \vec{a} = \vec{b} \times \vec{r}$, $\vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3$ and

$\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1, \alpha \in \mathbb{R}$, then the value of

$\alpha + |\vec{r}|^2$ is equal to :

- (1) 9
- (2) 15
- (3) 13
- (4) 11

Q3: 16 March (Shift 2) - Numerical

Let $\vec{c}$ be a vector perpendicular to the vectors

$\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$

If $\vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = 8$ then the value of $\vec{c} \cdot (\vec{a} \times \vec{b})$

is equal to _____

Q4: 17 March (Shift 1) - Single Correct

Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 7\hat{i} + \hat{j} - 6\hat{k}$.

If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}$, $\vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -3$, then $\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k})$

Questions with Answer Keys

MathonGo

is equal to :

(1) 12

(2) 8

(3) 13

(4) 10

Q5: 17 March (Shift 1) - Numerical

If $\vec{a} = \alpha\hat{i} + \beta\hat{j} + 3\hat{k}$

$\vec{b} = -\beta\hat{i} - \alpha\hat{j} - \hat{k}$ and

$\vec{c} = \hat{i} - 2\hat{j} - \hat{k}$

such that $\vec{a} \cdot \vec{b} = 1$ and $\vec{b} \cdot \vec{c} = -3$, then

$\frac{1}{3}((\vec{a} \times \vec{b}) \cdot \vec{c})$ is equal to _____

Q6: 17 March (Shift 2) - Single Correct

Let O be the origin. Let $\vec{OP} = x\hat{i} + y\hat{j} - \hat{k}$ and $\vec{OQ} = -\hat{i} + 2\hat{j} + 3x\hat{k}$, $x, y \in \mathbb{R}$, $x > 0$, be such that

$|\vec{PQ}| = \sqrt{20}$ and the vector $\vec{OP}$ is perpendicular to $\vec{OQ}$. If $\vec{OR} = 3\hat{i} + z\hat{j} - 7\hat{k}$, $z \in \mathbb{R}$, is coplanar with $\vec{OP}$ and $\vec{OQ}$, then the value of $x^2 + y^2 + z^2$ is equal to

(1) 7

(2) 9

(3) 2

(4) 1

Q7: 17 March (Shift 2) - Numerical

Let $\vec{x}$ be a vector in the plane containing vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$. If the vector $\vec{x}$ is perpendicular to $(3\hat{i} + 2\hat{j} - \hat{k})$ and its projection on $\vec{a}$ is $\frac{17\sqrt{6}}{2}$, then the value of $|\vec{x}|^2$ is equal to _____.

Questions with Answer Keys

MathonGo

Q8: 18 March (Shift 1) - Single Correct

A vector $\mathbf{a}$ has components $3p$ and 1 with respect to a rectangular cartesian system. This system is rotated through a certain angle about the origin in the counter clockwise sense. If, with respect to new system, $\mathbf{a}$ has components $p + 1$ and $\sqrt{10}$, then a value of p is equal to:

- (1) 1
- (2) $-\frac{5}{4}$
- (3) $\frac{4}{5}$
- (4) -1

Q9: 18 March (Shift 2) - Single Correct

In a triangle ABC, if $|\vec{BC}| = 8$, $|\vec{CA}| = 7$, $|\vec{AB}| = 10$, then the projection of the vector $\vec{AB}$ on $\vec{AC}$ is equal to:

- (1) $\frac{25}{4}$
- (2) $\frac{85}{14}$
- (3) $\frac{127}{20}$
- (4) $\frac{115}{16}$

Q10: 18 March (Shift 2) - Single Correct

Let $\vec{a}$ and $\vec{b}$ be two non-zero vectors perpendicular to each other and $|\vec{a}| = |\vec{b}|$. If $|\vec{a} \times \vec{b}| = |\vec{a}|$, then the angle between the vectors

$(\vec{a} + \vec{b} + (\vec{a} \times \vec{b}))$ and $\vec{a}$ is equal to:

- (1) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$
- (2) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$
- (3) $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$
- (4) $\sin^{-1}\left(\frac{1}{\sqrt{6}}\right)$

Answer Key

Q1 (1)

Q2 (2)

Q3 (28)

Q4 (1)

Q5 (2)

Q6 (2)

Q7 (486)

Q8 (4)

Q9 (2)

Q10 (2)